Reg. No: ……
Name: ……

**MAHATHMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025**
**THIRD SEMESTER**
**DSC A- MG3DSCBTG200- BIOPHYSICS AND INSTRUMENTATION**
(2024 ADMISSION ONWARDS)

**Duration: 1½** **Maximum Marks: 50**

**Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

**Part A**

Answer any **10** questions
Each question carries **2** marks **(10×2= 20)**

1. Define enthalpy **[K]/ [CO1]**
2. Name the laws of thermodynamics. **[U]/ [CO3]**
3. What is Osmosis? **[U]/ [CO2]**
4. Which law is applied in the working of colorimeter? **[K]/ [CO3]**
5. Which bond is formed between amino acids in a protein? **[K]/ [CO3]**
6. Name the different structural forms of DNA? **[U]/ [CO2]**
7. Define radioactivity.? **[K]/ [CO2]**
8. Name the parts of a compound microscope. **[K]/ [CO1]**
9. State the principle of ion exchange chromatography. **[U]/ [CO1]**
10. Define Biosensors. **[K]/ [CO2]**
11. Who invented the electrophoresis technique? **[K]/ [CO2]**
12. State the principle of HPLC. **[U]/ [CO3]**

**Part B**

Answer any **5** questions
Each question carries 4 marks **(5×4= 20)**

13. Describe the structure and components of electron microscope. **[K]/ [CO1]**
14. Explain the Laws of Thermodynamics. **[K]/ [CO1]**
15. Compare and Contrast SEM and TEM. **[An]/ [CO2]**
16. Describe the principle and applications of biosensors. **[A]/ [CO2]**
17. Explain the applications of autoradiography in molecular biology. **[A]/[CO3]**
18. Describe the role of water in living organisms. **[K]/ [CO3]**
19. Define polarization and Interference of light. Mention its significance in microscopy. **[U]/ [CO3]**

**Part C**

Answer any **1** question
Each question carries **10** marks **(1×10= 10)**

20. Describe the principle, construction, working and applications of GM counter. **[K]/ [CO1]**
21. Explain the formation of peptide bonds and the significance of the Ramachandran plot in understanding protein structure. **[U]/[CO3]**